

EDUCATION

Harvey Mudd College, Claremont, California
Bachelor of Science of Engineering (5x Dean's List)
Dacia Toll Scholarship Recipient

Expected May 2026

RELEVANT COURSEWORK

In Progress: Compressible Flow, Mechanical Design, Manufacturing Planning/Execution
Completed: Multivariable Calculus, Data / Prob. / Stats for Engr., Diff. Equations / Applied Math for Engr., Thermodynamics, Elec. & Magnetic Circuits / Devices, Materials Engr., Experimental Engr., Intro to Engr. Design & Manufacturing, Advanced Systems Engr., Continuum Mechanics, Computer Science, Mechanics & Wave Motion, Fluid Mechanics

SKILLS

CAD/CAE Software: SolidWorks, AutoCAD, Revit, COMSOL Multiphysics
Programming & Data Analysis: MATLAB, Python, LabView, R, C, SystemVerilog, Excel, VS Code
Fabrication & Prototyping: Advanced 3D Printing (SLA, SLS), Laser Cutting, Milling, Lathe Operation, Soldering Programming
Language: Python, Javascript, C, Arduino, SystemVerilog, R, Excel
Languages: English (Native), Spanish (Native), and French (Limited Proficiency)

EXPERIENCE

Blue Origin, Claremont, California

August 2025 - Present

Mechanical Team Lead, Clinic Project - Automating ground-side umbilical mating procedure for New Glenn rocket to enhance alignment precision, operational speed, and personnel safety.

- Lead mechanical sub-team in devising and designing a novel actuation system to autonomously guide and mate large umbilical panels, addressing critical mission requirements for millimeter alignment accuracy and efficiency.
- Drive mechanical design process from concept to detailed modeling, conducting trade studies on potential mechanisms and performing structural analysis (FEA) in CAD.
- Collaborate closely with controls and sensor sub-teams to define system architecture, select hardware, and assure seamless integration of mechanical system with a master control algorithm.

Flow Imaging Lab Research, Claremont, California

August 2024 - Present

Student Researcher | CAD Lead

- Inspired by hydrodynamics of archer fish, investigating novel propulsion systems for aerial-aquatic robots by collecting data across a parameter set of varying amplitude, frequency, tow speed, fin stiffness, and fraction submerged.
- Lead design and fabrication of experimental apparatus and mounting fixtures using SolidWorks, serving as lab's CAD Manager.
- Manage and calibrate a 6-axis load cell to precisely quantify hydrodynamic forces and torques generated during water-exit events.
- Analyze fluid dynamics in Matlab by implementing Particle Image Velocimetry (PIV) to visualize flow patterns and utilizing Computational Fluid Dynamics (CFD) to validate experimental data and inform trajectory design.
- Maintain and optimize all data acquisition equipment and experimental tank to assure operational readiness and data integrity.

Mudd Rocketry Club, Claremont, California

September 2022 - Present

Mechanical Engineer

- Launched and recovered Gladius-III reaching 17172 feet with 99.95% altitude accuracy.
- Fabricated composite bulkheads, centering rings, and fins to tight tolerances using a mill, waterjet cutter, laser cutter, router, rotary tool, drill, and drivers.
- Collaborated closely with the Avionics and Recovery sub-teams to ensure seamless integration of components and systems.

Indomo TX, Claremont California

January 2025 - May 2025

Mechanical Engineer - Redesigned a next-generation autoinjector for a clinical-stage company's novel, at-home acne treatment.

- Engineered key mechanical improvements for device redesign, focusing on enhancing user experience, ergonomics, and design for manufacturability (DFM).
- Implemented a critical safety interlock integrating user activation mechanism with a skin-contact sensor, eliminating risk of accidental device discharge.
- Developed a decoupled mechanism to separate needle deployment from drug delivery, optimizing system's mechanical timing to enhance user safety and ensure reliable operation.
- Engineered complex CAD models and validated subsystem designs through rapid 3D prototyping, ensuring higher reliability and faster development.